

# **Can Machine Learning Variable Selection and Individual Commodity Fundamentals Improve Futures Return Forecasting?**

Leo Specks

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# Contents

|          |                                                                 |          |
|----------|-----------------------------------------------------------------|----------|
| <b>1</b> | <b>Introduction</b>                                             | <b>3</b> |
| 1.1      | Motivation & Background . . . . .                               | 3        |
| 1.2      | The Research Gap . . . . .                                      | 3        |
| 1.3      | Research Questions & Hypotheses . . . . .                       | 3        |
| 1.4      | Contributions to the Literature . . . . .                       | 3        |
| 1.5      | Thesis Outline . . . . .                                        | 3        |
| <b>2</b> | <b>Literature Review</b>                                        | <b>4</b> |
| 2.1      | Theories of Commodity Futures Returns . . . . .                 | 4        |
| 2.2      | Aggregate Commodity-Specific Factors . . . . .                  | 4        |
| 2.3      | Individual Commodity Fundamentals . . . . .                     | 4        |
| 2.4      | Macroeconomic Predictors . . . . .                              | 4        |
| 2.5      | ML Methods in Return Forecasting . . . . .                      | 4        |
| 2.6      | Volatility Regimes and Predictability . . . . .                 | 4        |
| <b>3</b> | <b>Methodology</b>                                              | <b>5</b> |
| 3.1      | Baseline Framework (Replication) . . . . .                      | 5        |
| 3.2      | Extension I: Penalized Regression . . . . .                     | 5        |
| 3.3      | Extension II: Individual Fundamental Predictors . . . . .       | 5        |
| 3.4      | OOS Evaluation Criteria . . . . .                               | 5        |
| 3.5      | Portfolio Back-Testing Framework . . . . .                      | 5        |
| <b>4</b> | <b>Data</b>                                                     | <b>6</b> |
| 4.1      | Commodity Futures Returns . . . . .                             | 6        |
| 4.2      | Macroeconomic Factors . . . . .                                 | 6        |
| 4.3      | Aggregate Commodity-Specific Factors . . . . .                  | 6        |
| 4.4      | Individual Fundamental Predictors . . . . .                     | 6        |
| 4.5      | Sample Splits and Descriptive Statistics . . . . .              | 6        |
| <b>5</b> | <b>Replication &amp; Baseline Results</b>                       | <b>7</b> |
| 5.1      | In-Sample Estimation (1989-2003) . . . . .                      | 7        |
| 5.2      | OOS Predictive Accuracy (2004-2012) . . . . .                   | 7        |
| 5.3      | Portfolio Back-Testing (2004-2012) . . . . .                    | 7        |
| 5.4      | Replication Assessment . . . . .                                | 7        |
| <b>6</b> | <b>Extension I: ML Variable Selection</b>                       | <b>8</b> |
| 6.1      | Model Specification and Implementation . . . . .                | 8        |
| 6.2      | Variable Selection Patterns . . . . .                           | 8        |
| 6.3      | OOS Forecasting Results - Original Sample (2004-2012) . . . . . | 8        |
| 6.4      | OOS Forecasting Results - Extended Sample (2013-2024) . . . . . | 8        |
| 6.5      | Portfolio Back-Testing . . . . .                                | 8        |
| 6.6      | Robustness Checks . . . . .                                     | 8        |

|           |                                                                   |           |
|-----------|-------------------------------------------------------------------|-----------|
| <b>7</b>  | <b>Extension II: Individual Commodity Fundamentals</b>            | <b>9</b>  |
| 7.1       | Predictor Construction by Commodity Group . . . . .               | 9         |
| 7.2       | Individual vs. Aggregate Factors: Are They Substitutes? . . . . . | 9         |
| 7.3       | OOS Forecasting Results (2004-2024) . . . . .                     | 9         |
| 7.4       | Combined Model . . . . .                                          | 9         |
| 7.5       | Portfolio Back-Testing . . . . .                                  | 9         |
| 7.6       | Regime Interaction . . . . .                                      | 9         |
| <b>8</b>  | <b>Synthesis &amp; Discussion</b>                                 | <b>10</b> |
| 8.1       | Model Horse Race . . . . .                                        | 10        |
| 8.2       | Statistical vs. Economic Loss Functions Revisited . . . . .       | 10        |
| 8.3       | Implications for Commodity Market Segmentation . . . . .          | 10        |
| 8.4       | Practical Implications . . . . .                                  | 10        |
| <b>9</b>  | <b>Conclusions</b>                                                | <b>11</b> |
| 9.1       | Summary of Findings . . . . .                                     | 11        |
| 9.2       | Limitations . . . . .                                             | 11        |
| 9.3       | Future Research Directions . . . . .                              | 11        |
| <b>10</b> | <b>Test Chapter</b>                                               | <b>12</b> |
| 10.1      | (Sub)plots and Tables . . . . .                                   | 12        |
| 10.2      | Citations . . . . .                                               | 14        |
| 10.3      | Codeblock . . . . .                                               | 14        |
|           | <b>References</b>                                                 | <b>16</b> |
|           | <b>Appendix</b>                                                   | <b>17</b> |

# **1 Introduction**

## **1.1 Motivation & Background**

## **1.2 The Research Gap**

## **1.3 Research Questions & Hypotheses**

## **1.4 Contributions to the Literature**

## **1.5 Thesis Outline**

## **2 Literature Review**

### **2.1 Theories of Commodity Futures Returns**

### **2.2 Aggregate Commodity-Specific Factors**

### **2.3 Individual Commodity Fundamentals**

### **2.4 Macroeconomic Predictors**

### **2.5 ML Methods in Return Forecasting**

### **2.6 Volatility Regimes and Predictability**

## **3 Methodology**

### **3.1 Baseline Framework (Replication)**

### **3.2 Extension I: Penalized Regression**

### **3.3 Extension II: Individual Fundamental Predictors**

### **3.4 OOS Evaluation Criteria**

### **3.5 Portfolio Back-Testing Framework**

## **4 Data**

### **4.1 Commodity Futures Returns**

### **4.2 Macroeconomic Factors**

### **4.3 Aggregate Commodity-Specific Factors**

### **4.4 Individual Fundamental Predictors**

### **4.5 Sample Splits and Descriptive Statistics**

## **5 Replication & Baseline Results**

### **5.1 In-Sample Estimation (1989-2003)**

### **5.2 OOS Predictive Accuracy (2004-2012)**

### **5.3 Portfolio Back-Testing (2004-2012)**

### **5.4 Replication Assessment**

## **6 Extension I: ML Variable Selection**

### **6.1 Model Specification and Implementation**

### **6.2 Variable Selection Patterns**

### **6.3 OOS Forecasting Results - Original Sample (2004-2012)**

### **6.4 OOS Forecasting Results - Extended Sample (2013-2024)**

### **6.5 Portfolio Back-Testing**

### **6.6 Robustness Checks**

## **7 Extension II: Individual Commodity Fundamentals**

### **7.1 Predictor Construction by Commodity Group**

### **7.2 Individual vs. Aggregate Factors: Are They Substitutes?**

### **7.3 OOS Forecasting Results (2004-2024)**

### **7.4 Combined Model**

### **7.5 Portfolio Back-Testing**

### **7.6 Regime Interaction**

## **8 Synthesis & Discussion**

### **8.1 Model Horse Race**

### **8.2 Statistical vs. Economic Loss Functions Revisited**

### **8.3 Implications for Commodity Market Segmentation**

### **8.4 Practical Implications**

## **9 Conclusions**

### **9.1 Summary of Findings**

### **9.2 Limitations**

### **9.3 Future Research Directions**

## 10 Test Chapter

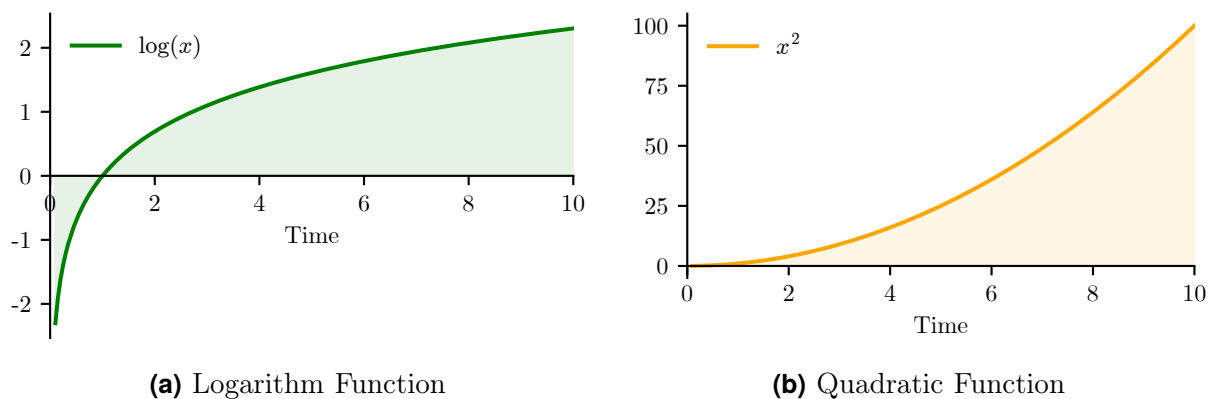
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**Figure 1**

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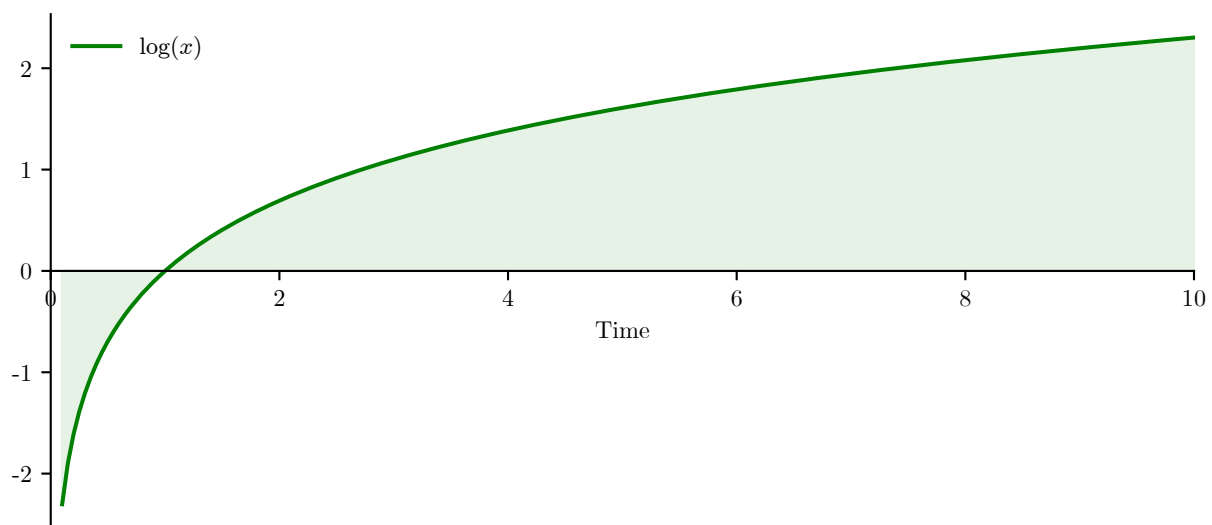


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## Figure 2

Example Figure (width should be 16cm)



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**Table 1**

UMD Factor: Summary Statistics (Jan 1927–Feb 2026)

| Statistic              | Monthly               | Annualized |
|------------------------|-----------------------|------------|
| Mean Return (%)        | 0.612                 | 7.34       |
| Standard Deviation (%) | 4.68                  | 16.21      |
| Sharpe Ratio           | -                     | 0.453      |
| t-statistic            | 4.51                  | -          |
| p-value (two-tailed)   | $7.11 \times 10^{-6}$ | -          |
| Number of Months       | 1,190                 | -          |

tincidunt eget. Praesent dictum sed eros in pharetra. Vivamus nec dictum risus. Integer ac massa at magna rutrum pulvinar eget nec nunc. Curabitur tempor at lorem vitae venenatis. Suspendisse auctor posuere iaculis. Suspendisse egestas lobortis diam. Integer mattis. The approach by Guidolin and Pedio (2021) has shown that this is realistic<sup>1</sup>.

## 10.2 Citations

This is a cite: Guidolin and Pedio, 2021, this a textcite: Guidolin and Pedio (2021), this a parencite: (Guidolin & Pedio, 2021), this a citeauthor: Guidolin and Pedio, this a citeyear: 2021, and this a parencite with two references: (Giampietro et al., 2018; Guidolin & Pedio, 2021).

**Table 2**

Citation Styles with BibLaTeX

| Command                    | authoryear     | numeric           |
|----------------------------|----------------|-------------------|
| <code>\cite{}</code>       | Author, Year   | [number]          |
| <code>\textcite{}</code>   | Author (Year)  | Author [number]   |
| <code>\parencite{}</code>  | (Author, Year) | [number]          |
| <code>\citeauthor{}</code> | Author         | Author            |
| <code>\citeyear{}</code>   | Year           | Year (not linked) |

Use `\parencite{}` at the end of a statement and `\textcite{}` when referencing in the middle of a sentence.

## 10.3 Codeblock

The algorithm can be viewed in Codeblock 1. Nullam egestas viverra purus vitae varius. Sed maximus tempor vestibulum. Nunc tempor nulla justo, a posuere lorem tincidunt eget. Praesent dictum sed eros in pharetra. Vivamus nec dictum risus. Integer ac massa at magna rutrum pulvinar eget nec nunc. Curabitur tempor at lorem vitae venenatis. Suspendisse auctor posuere iaculis. Suspendisse egestas lobortis diam. Integer mattis.

<sup>1</sup>This would be the first footnote.

**Code 1**

Example Codeblock

```
# KPI OVERVIEW
kpis = pd.DataFrame({
    "Ann. Return (approx)": ann_return_simple,
    "Ann. Return (compounding)": ann_return,
    "Ann. Volatility": ann_vol,
    "Sharpe": sharpe,
    "CAGR": cagr,
    "Max Drawdown": max_dd
})

kpis.T
```

---

## References

- Giampietro, M., Guidolin, M., & Pedio, M. (2018). Estimating stochastic discount factor models with hidden regimes: Applications to commodity pricing. *European Journal of Operational Research*, 265(2), 685–702. <https://doi.org/10.1016/j.ejor.2017.07.045>
- Guidolin, M., & Pedio, M. (2021). Forecasting commodity futures returns with stepwise regressions: Do commodity-specific factors help? *Annals of Operations Research*, 299(1–2), 1317–1356. <https://doi.org/10.1007/s10479-020-03515-w>
- Auer, B. R. (2021). Have trend-following signals in commodity futures markets become less reliable in recent years? *Financial Markets and Portfolio Management*, 35(4), 533–553. <https://doi.org/10.1007/s11408-021-00385-5>
- Guidolin, M., & Ionta, S. (2026). Predicting commodity returns with climate variables: Statistical loss functions vs. economic value. *Economics Letters*, 265, 113028. <https://doi.org/10.1016/j.econlet.2026.113028>

\*not cited yet

## **Appendix**